

Praneel Baskar

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EDUCATION

University of Illinois at Urbana-Champaign (UIUC)

Expected May 2028

Bachelor of Science in Electrical Engineering

- **GPA:** 3.95/4.00 | **Dean's List** | **James Scholar Honors Program**
- **Relevant Coursework:** Computer Systems & Programming, Fundamentals of Robotics, Analog Signal Processing, Discrete Math, Linear Algebra w/ Comp. Applications.

SKILLS

Software: Python, C, Java, Git/GitHub, Flask, FastAPI, SQLite, Unit & Integration Testing, Technical Documentation.

ML & Perception: PyTorch, OpenCV, CoreML, Vision-Language Models, On-Device Inference, Imitation Learning.

Systems & Hardware: FSM / Digital Logic Design, Sensor Integration, Closed-Loop Control, Oscilloscope Debugging, Analog Signal Processing.

EXPERIENCES

Trainify Labs

May 2026 – Present

Computer Vision Intern

- Built an end-to-end **on-device perception pipeline** for a production iOS app, converting a pose-detection model to **CoreML** for real-time inference on the Apple Neural Engine.
- Diagnosed and fixed a multi-person tracking failure by replacing confidence-based matching with **appearance-based re-identification**, shipping a fix validated on previously failing footage.

SecuraAI

May 2026 – Aug 2026

Lead Software Engineering Intern

- **Led a team of interns** building a **Python CLI framework** that tests vision model reliability across multiple providers through one config-driven interface.
- Root-caused a model failure to a **vision-encoder downsampling threshold** rather than model behavior, then validated the fix through **controlled replay** against archived data.

First Tech Challenge (Robotics)

Aug 2021 – March 2025

Hardware Team Lead

- Implemented autonomous navigation logic in **Java**, integrating encoder and gyroscope feedback into **closed-loop control** for real-time field positioning.
- Engineered a robotic intake/delivery system with custom gear ratios and 3D-printed parts, cutting cycle time by **60%** and reaching **World Semi-Finals**.

PROJECTS

Visual Agent - Agentic Computer-Vision System

May – June 2026

- Built a **perceive-reason-act loop** (LangGraph, OpenCV, SQLite) that captions webcam frames with **Qwen2-VL-7B** and triggers desktop alerts on detected conditions.
- Served the model on a remote **H100** blocked behind a university firewall by running **Flask** in a background thread over an ngrok tunnel, streaming base64-JPEG frames each cycle.
- Root-caused false alerts by querying the caption log, finding **58 of 78** phone captions were denials, then split a shared negation guard that had silently suppressed real alerts.

DriveSafe - Multi-Sensor Safety System, Most Commercializable Award

Jan – May 2026

- Designed a three-state **finite state machine** in discrete logic with no microcontroller, scaling its alarm tolerance with vehicle speed through a 4:1 multiplexer.
- Fixed false timeouts by **synchronizing** an asynchronous fault signal through a D flip-flop, and traced a dead comparator to an ungrounded reference pin using a scope and datasheet.

Bracket Bot - Robot Learning, UIUC ACM SIGRobotics

Dec 2025 – Present

- Collecting teleoperated demonstrations on dual **SO-100 arms** to train a cloth-folding **imitation-learning** policy, within a team project on self-supervised visual world models.